

THE VILLAGES, FL, USA

WMO: 722012

Lat: 28.950N

Lon: 81.967W

Elev: 27

StdP: 101.00

Time Zone: -5.00 (NAE)

Period: 02-18

WBAN: 92817

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.0	2.6	-7.6	2.0	7.3	-4.9	2.5	8.8	8.8	18.2	8.1	17.7	1.3	0	0.336

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	34.0	25.1	33.0	24.9	32.6	24.8	27.9	29.7	27.3	29.6	26.7	29.3	2.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.5	23.4	28.4	27.0	22.7	28.1	26.1	21.6	27.7	89.6	30.1	85.9	29.1	83.4	30.1	34.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.9	7.0	5.8	-1.5	35.5	1.4	0.8	-2.5	36.0	-3.3	36.5	-4.1	37.0	-5.1	37.6		
			-3.7	28.9	1.6	2.2	-4.8	30.4	-5.7	31.7	-6.6	32.9	-7.8	34.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.9	14.2	15.5	18.4	21.5	24.7	27.1	27.8	27.8	26.8	23.1	18.8	16.2	
	DBStd	5.90	5.03	4.66	4.03	3.18	2.35	1.56	1.21	1.27	1.29	3.32	3.88	4.94	
	HDD10.0	38	19	9	1	0	0	0	0	0	0	0	1	8	
	HDD18.3	453	144	100	50	11	0	0	0	0	0	7	40	100	
	CDD10.0	4364	150	163	262	344	457	512	552	551	503	405	265	201	
	CDD18.3	1738	17	21	53	106	199	262	293	292	253	154	54	34	
	CDH23.3	15140	99	151	417	916	1859	2347	2832	2746	2147	1098	331	198	
	CDH26.7	5722	8	18	80	265	767	1012	1228	1139	819	312	51	23	
Wind	WSAvg	2.2	2.3	2.5	2.6	2.6	2.5	2.1	1.6	1.7	1.9	2.0	2.0	2.1	
Precipitation	PrecAvg	1248	70	60	92	61	78	192	165	189	159	82	41	58	
	PrecMax	1563	170	148	222	191	364	370	256	355	389	179	114	215	
	PrecMin	753	10	13	0	10	11	26	83	113	54	16	9	10	
	PrecStd	203	46	39	67	41	76	83	49	59	90	49	29	52	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.6	28.7	30.2	32.5	34.0	35.1	35.2	34.9	33.0	32.2	30.0	28.7	
		MCWB	21.8	21.8	21.1	22.4	22.2	24.7	25.6	26.1	25.0	25.1	25.0	25.6	
	2%	DB	26.2	27.2	28.8	31.1	32.8	33.8	33.9	33.8	32.5	31.0	27.8	27.3	
		MCWB	20.3	20.1	20.4	21.6	22.3	24.6	25.3	26.2	24.8	24.0	22.3	22.4	
	5%	DB	24.0	25.9	27.5	29.1	32.2	32.7	32.9	32.8	32.1	29.2	27.1	26.0	
		MCWB	19.0	19.3	19.5	20.7	22.0	24.5	25.2	26.0	24.7	23.3	21.1	21.2	
	10%	DB	22.4	23.0	26.1	27.9	31.1	32.2	32.4	32.3	31.1	28.0	25.3	23.8	
		MCWB	18.4	17.8	18.7	19.8	22.0	24.5	25.2	26.0	24.5	22.4	20.0	20.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.4	24.0	23.5	26.2	25.7	27.6	27.9	32.7	28.7	27.4	25.9	26.4
			MCDB	25.9	26.1	26.4	27.5	29.2	30.1	30.0	32.7	30.4	29.2	28.2	27.7
2%		WB	22.1	21.9	22.4	23.7	24.7	27.0	27.3	28.4	27.4	26.0	24.2	24.1	
		MCDB	24.0	24.6	25.8	27.5	29.1	29.9	29.8	29.6	28.9	27.8	26.3	25.1	
5%		WB	20.6	20.8	21.3	22.6	24.0	26.2	26.7	27.4	26.4	25.1	22.8	22.4	
		MCDB	23.2	23.8	24.8	26.8	28.5	29.6	29.7	29.8	28.4	27.6	24.9	24.5	
10%		WB	19.1	19.4	20.4	21.7	23.3	25.6	26.1	26.7	25.8	24.2	21.7	21.1	
		MCDB	21.8	22.3	23.8	26.0	27.5	29.4	29.5	29.5	28.5	26.9	23.9	23.3	
Mean Daily Temperature Range	5% DB	MDBR	11.9	12.3	12.4	12.3	11.7	9.5	9.3	8.9	8.9	10.1	11.3	11.5	
		MCDBR	12.1	13.0	12.9	12.9	12.6	10.6	10.0	9.7	9.7	9.9	11.1	11.2	
		MCWBR	7.2	6.8	5.3	4.7	3.8	3.6	3.4	3.9	3.6	4.2	5.8	6.6	
	5% WB	MCDDBR	11.0	11.3	10.9	11.0	10.6	8.9	9.0	8.7	8.3	8.1	9.1	10.2	
		MCWBR	7.5	7.5	5.6	5.5	3.9	3.9	3.8	4.6	4.0	4.2	5.6	7.3	
Clear-Sky Solar Irradiance	taub	0.337	0.350	0.367	0.388	0.426	0.456	0.485	0.468	0.420	0.392	0.354	0.341		
	taud	2.519	2.491	2.439	2.356	2.301	2.289	2.208	2.270	2.390	2.437	2.494	2.530		
	Ebn at Noon	903	920	922	907	869	838	812	825	857	862	876	881		
	Edh at Noon	90	100	112	125	133	134	145	135	116	104	91	85		
All-Sky Solar Radiation	RadAvg	3.37	4.01	5.13	6.15	6.46	5.79	5.61	5.37	4.86	4.43	3.64	3.08		
	RadStd	0.24	0.32	0.43	0.36	0.41	0.47	0.37	0.36	0.33	0.32	0.20	0.26		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (44 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	-6	-52	+159	+115

Nomenclature: See separate page